BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ABBOTTABAD

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2022

PHYSICS

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 12

Note: Attempt all questions of Section-A by filling the corresponding bubble on the MCQs RESPONSE SHEET. It is mandatory to return the attempted MCQs sheet to the Superintendent within given time.

Q1. If the turns ratio is 5, the secondary voltage is greater than the primary voltage by a factor of .....

(A) 0.5
(B) 0.3
(C) 0.8
(D) 0.2

Q2. The Boolean equation for OR gate is .....

(A) $X=A\cdot\bar{B}$
(B) $X=AB$
(C) $X=A+\bar{B}$
(D) $X=A+B$

Q3. Electric (or) electromagnetic representations of data (or) information is termed as .....

(A) transmission
(B) signal
(C) software
(D) computer

Q4. Which derived unit is equivalent to tesla T?

(A) N/Am
(B) NA/m
(C) Nm/A
(D) Am/N

Q5. When the current through a wire is 2.0 A and voltage across its ends is 3.0 V then resistance is ..... Ω

(A) 0.6
(B) 1.5
(C) 2.5
(D) 5

Q6. What is the number of neutrons in the plutonium $^{242}PU_{92}$?

(A) 92 (B) 134

(C) 242 (D) 150

Q7. A shock absorber is designed to introduce ..... forces.

(A) electric (B) damping

(C) magnetic (D) gravitational

Q8. A reflected sound that arrives before 0.10 s is perceived as an increase in volume is called a/an .....

(A) echo (B) pitch

(C) reverberation (D) quality

Q9. The focal length of a convex mirror with radius of curvature 10 cm is ..... cm.

(A) −5 (B) −10

(C) +5 (D) +10

Q10. The human eye forms the image of an object at its .....

(A) iris (B) retina

(C) cornea (D) pupil

Q11. Four identical 1μF capacitors are connected together electrically. What is the least possible capacitance of the combination?

(A) 1μF (B) 4μF

(C) 1/4μF (D) 8μF

Q12. The current which reverses direction after regular interval of time is .....

(A) DC (B) AC

(C) conventional (D) both DC and AC

SECTION-B

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 32

Q1. Attempt any eight of the following short questions. Each question carries 4 marks

(i) Is the restoring force on a mass attached to spring in simple harmonic motion ever zero? If so, where?

(ii) Why is the intensity of an echo less than that of the original sound?

(iii) What is the speed of sound in air at −20°C?

(iv) What are optical fibres? Give three applications of optical fibres.

(v) State Joule's law and write down its mathematical form.

(vi) The sum of the charges on both plates of a capacitor is zero. What does a capacitor store?

(vii) Explain why is it possible for birds to perch safely on high tension wires without being electrocuted?

(viii) The primary coil of a transformer is connected to a DC battery. Is there an emf induced in the secondary coil. Why?

(ix) Define logic gates. Describe the operation of NAND and NOR logic gates by drawing truth tables.

(x) Can internet be used for shopping? Give an example.

(xi) The atomic number of one particular isotope is equal to its mass number. Which isotope is it?

SECTION-C

Maximum Marks: 21

> Note: Attempt any three of the following questions. Each question carries 7 marks

Q2.

(i) Show that mass spring system executes simple harmonic motion.

(ii) How is sound reflected? Describe the difference between echo and reverberation.

Q3.

(i) What is meant by refraction of light? What is the index of refraction? State and explain laws of refraction with diagrams.

(ii) Gulalai is viewing a flea using a magnifier with f=3.0 cm. If her near point is at N=25 cm then calculate the maximum magnification she can get.

Q4.

(i) State and explain Coulomb's law.

(ii) What is potential difference and emf? Differentiate between them.

Q5.

(i) Describe the working principle of cathode ray oscilloscope (CRO) and make list of its uses.

(ii) You have a 100.0 µg sample of polonium-218. It decays by alpha emission with a half-life of 3.0 min. How much of the pure sample is left after 9.0 min?